Self-Service Coffee Kiosk System — UML Component Diagram

Layered Architectural Pattern | PES University Software Engineering Lab 3 (UE24CS252AA)

Presentation Layer

<<component>>

User Interface Component

Touch Screen Interaction | Menu Selection | Size & Type Select

«interface» Order Interface

selectCoffee(type, size), calculateTotal(), submitOrder()

Business Layer

<<component>>

Order Manager Component

Order Lifecycle Orchestration | Price Calculation | Transaction Coordination

«interface» Payment Interface

processPayment(), authorizeCard() [PCI SDK]

<<component>>

Payment Service Component

EMV Card Terminal Bridge | PCI Tokenization | Credit Card Only

«interface» Receipt Interface

printReceipt(), getPrinterStatus() [ESC/POS]

<<component>>

Receipt Printer Component

ESC/POS Driver | Itemized Slip Formatter | Hardware Status

«interface» Database Interface

getMenuData(), getPricing(), saveOrderRecord() [SQL / JDBC]

Data Layer

<<component>>

Database Component

Menu & Pricing Catalog | Order Transaction Store | Local SQLite DB

Provided Interface (Ball) — **Required Interface (Socket)**

Coffee Types: Espresso, Americano, Latte | Sizes: Small, Large | Payment: Credit Card | Hardware: Receipt Printer, Touch Screen